

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
7th Semester of B. Pharm. Examination
Theory University Examination November/ December 2017
PH418 Medicinal Chemistry-III

Date: 24.11.2017, Friday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. **SECTION – I** comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to **SECTION – I** are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for **SECTION – I** is 30 minutes. Answers to **SECTION – I** must be written during the first 30 minutes of the examination.
3. Answers to **SECTION – II** and **SECTION – III** are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION – I

Q 1 Attempt all questions. Each question is of one mark. 20

1. 1, 3-Diamino cyclohexane central ring absent in _____.
[A] Kanamycin
[B] Gentamicin
[C] Tobramycin
[D] Streptomycin
2. _____ is 8-aminoquinoline antimalarial agents.
[A] Pyrimethamine
[B] Primaquine
[C] Mefloquine
[D] Proguanil
3. Streptomycin is obtained from _____.
[A] *Micromonospora purpurea*
[B] *Streptomyces griseus*
[C] *Streptomyces rimosus*
[D] *Streptomyces tenebrarius*
4. Identify the folic acid analogue as an anticancer agents.
[A] Cladribine
[B] Fludarabine
[C] Methotrexate
[D] Azathioprine
5. Which of the following is Non-nucleoside reverse transcriptase inhibitor.
[A] Efavirenz
[B] Abacavir
[C] Zidovudine
[D] Zalcitabine

6. Enfuvirtide is _____.
 - [A] CCR5 Receptor Inhibitor
 - [B] Fusion Inhibitor
 - [C] Integrase Inhibitor
 - [D] Protease Inhibitor
7. Tamoxifen is a _____.
 - [A] Antiandrogen
 - [B] Aromatase inhibitor
 - [C] Selective estrogen receptor modulators
 - [D] Glucocorticoids
8. _____ is triazole containing antifungal agent.
 - [A] Miconazole
 - [B] Itraconazole
 - [C] Ketoconazole
 - [D] Clotrimazole
9. All are correct about Green Chemistry. Except One
 - [A] Using Non-Toxic Reagents
 - [B] Maximum Use of Organic Solvent
 - [C] Use of Catalysts in place of Reagents
 - [D] Waste Minimisation at Source
10. Terbinafin inhibits _____.
 - [A] Conversion of squalene to lanosterol
 - [B] Fungal cell wall
 - [C] Glucan synthase
 - [D] Conversion of lanosterol to ergosterol
11. _____ isomer of Chloramphenicol is clinically active.
 - [A] D-threo
 - [B] L-threo
 - [C] L-erythro
 - [D] D-erythro
12. The naturally occurring tetracyclines contains which of the followings?
 - [A] α -C3-C4 keto enol group
 - [B] α -C3 Dihydroxy substituent
 - [C] α -C3 dimethyl amino substituent
 - [D] α -C4 dimethyl amino substituent
13. Which is following is not a prodrug
 - [A] Proguanil
 - [B] Prontosil red
 - [C] Acyclovir
 - [D] Trimethoprim
14. Clavunic acid has a beta lactam ring fused with
 - [A] Thienyl system
 - [B] Thiadiazole
 - [C] Thiazolidone
 - [D] Oxazolidone

15. Which of the following is not an alkylating agent?
- [A] Cyclophosphamide
 - [B] 6-Mercaptopurin
 - [C] Chlorambucil
 - [D] Thiotepa
16. Acrolein toxicity is associated with _____
- [A] Cyclophosphamide
 - [B] 6-Mercaptopurin
 - [C] Melphalan
 - [D] Dacarbazine
17. Which is inhibitor of sterol-14- α -demethylase?
- [A] Naftifine
 - [B] 5-fluocytosine
 - [C] Ciclopirox
 - [D] Ketoconazole
18. Match the following
- | MOA | Drugs |
|---|----------------|
| 1. Inhibit Arabinogalactan Synthase | a. Rifampin |
| 2. Inhibit DNA dependant RNA polymerase | b. Dapson |
| 3. Inhibit folic acid Synthesis | c. Ethambutol |
| 4. Mycolic acid synthesis | d. Isoniazide |
| | e. Ethionamide |
- [A] 1-c, 2-a, 3-b, 4-d
 - [B] 1-c, 2-a, 3-b, 4-e
 - [C] 1-c, 2-a, 3-e, 4-d
 - [D] 1-c, 2-a, 3-d, 4-b
19. Chemically chlorhexidine is _____ derivative.
- [A] Amide
 - [B] Anilide
 - [C] Biguanide
 - [D] Furan
20. Albendazole has _____ heterocyclic ring structure.
- [A] Piperazine
 - [B] Thiazole
 - [C] Benzimidazole
 - [D] Triazole

SECTION – II

- Q 2 Attempt any TWO of the following**
- A Describe SAR of Sulphonamides. 05
 - B Write a detail note on chemistry of aminoglycosides. 05
 - C What is β - Lactams? Describe SAR and mechanism of action of it. 05
- Q 3 Attempt any TWO of the following**
- A Classify antifungal agents with atleast one structure from each class. 05
 - B Classify antineoplastic agents with their chemical structures. (atleast one structure from each class). 05
 - C Draw a synthetic pathway for sulphamethoxazole and metronidazole. 05

SECTION – III

- Q 4 Attempt any FOUR of the following**
- A Explain SAR and mechanism action of Fluoroquinolones. 05
 - B List out various antiseptic and disinfectant with their structure. 05
 - C Discuss SAR of tetracyclines. 05
 - D Write a chemistry of isoniazide with mechanism of action. 05
 - E Write a SAR of azole containing antifungal agents. 05
 - F Give chemical classification of antiamebic agents. Explain mechanism of action of metronidazole. 05
- Q 5 Attempt any FOUR of the following**
- A Write a SAR and toxicities of chloramphenicol in details. 05
 - B Write a note on green chemistry and its principle. 05
 - C Classify antiviral agents with atleast one structure from each class. Give mechanism of action of acyclovir. 05
 - D Write a short note on macrolide antibiotics 05
 - E Classify cephalosporin derivatives with their structures. 05
 - F Write a synthetic pathway and mechanism of action of chloroquine. 05